TECHNICAL SPECIFICATION

Supply, installing , testing and Commissioning of 80 W LED lamps along with fixtures for switchyard lighting in Substations.

3.1 SCOPE:

This Specification covers design, manufacture, assembly, shop testing, forwarding and delivery F.O.R. destination including unloading at stores / site free installation and commissioning of LED out door light fittings as per the schedule .

3.2 STANDARDS:

Except otherwise specified or implied herein the LED Luminaries/ Light fittings shall comply in all respects with the applicable latest International/ Indian standards for the technical specifications specified here in..

3.3 TYPE AND RATING:

The LED luminaries should be suitable for yard lighting in substation switch yard application and should provide lumen output .

The Bidder shall furnish all valid type test reports along with the bid including the following as applicable as per the standards.

i. Total Luminous flux
ii. Luminous Intensity distribution
iii. Electrical power
iv. Luminous efficacy
v. Color characteristics
   - Chromaticity
   - CCT
   - CRI
vi. Degree of Protection IP-65

GUARANTEED TECHNICAL PARTICULARS REQUIREMENTS OF LED LIGHT FITTINGS/ LUMANAIRS FOR OUTDOOR YARD LIGHTING AT SUBSTATIONS

Supply Source	85-264 volt. However, conditions for voltage fluctuations should be considered and the system should be robust enough to withstand such variation in supply source.
Frequency	47 to 63 Hz
Luminaire Efficacy	The luminaries should have a system efficacy of greater than 90 lm/W
LED requirement	High power LED with correlated color temperature of 4000K to 6500K+/-500K.
Luminaire technical requirement	The LED luminaries should be suitable for substation outdoor yard lighting application and should provide similar lumen output to that of the existing light and uniformity.
Usage Hours	12 hrs per day
Power Consumption	Optimum for the efficacy and Lux equivalent to 250 W HPSV Luminaires.
Power Factor	Greater than 0.95
Life Expectancy	About 50,000 burning hours at a 35^0 C ambient temperature with 70% lumen maintenance.
Colour Rendering Index	Minimum CRI of 75
Control Circuit	Compatible with LED
Construction	Cast Aluminium with Toughened Glass sealed to IP 65 with SS Toggles.
Required illumination level	At main equipment 50 lux and at other locations 20 lux.
Warranty	3 years on entire luminary, illumination levels and on LEDs for 100% replacement.

GENERAL REQUIREMENTS FOR LED YARD LIGHTING

i. LED lamps of CREE/ NICHIA/OSRAM/SEOUL/PHILIPS LUMILEDS/ LEDNIUM shall be used for the purpose.

ii. The luminary casing/housing shall be of stainless sheet steel SS304 grade or aluminum having high conductivity preferably to grade 5000 or LM9 or similar to high conductivity heat sink material.

iii. The electronic components used shall be as follows:-

   a. IC (Integrated Circuit) shall be of industrial grade or above.

   b. Metallic film or paper or polyester capacitor shall be rated for a sustained operating temperature of 105^0 C.

   c. The resistors shall be preferably made of metal film of adequate rating. The actual rating versus loading shall be by a factor of 3.

   d. The junction temperature of the switching devices such as transistors or MOSFETs etc. shall not exceed 125^0 C (allowing thermal margin of 25^0C).

   e. The protective cum adhesive coating used on PCBs should be cleared and transparent and should not affect color code of electronic components or the product code of the company.

   f. The construction PCBs and the assembly for the components for the PCBs should be as per IS/IEC standards.

   g. The electronics covered for this equipment shall pass all the tests called for in the above specification. The tenderer shall indicate the deviation or compliance otherwise the offer shall not be considered for evaluation.

   h. The infrastructure of Quality Assurance facilities as called for in the above specification must be available at the manufacturing facility. In-house testing facility for Quality Assurance should be present. The compliance shall be indicated clearly in the tender itself.

iv. The connecting wires used inside the luminaire, shall be low smoke halogen free, fire radiant e-beam cable and use protection shall be provided in input side.

v. Care shall be taken in the design that there is no water stagnation anywhere. The entire housing shall be dust and water proof having IP65 protection as per IEC 60529.

vi. The LED module(s), Driver gear, etc. shall be designed in such a way so that temperature of heat sink shall not exceed 15^0C above the ambient temperature.

vii. All the materials used in the luminaire shall be halogen free and fire retardant confirming to safety standard UL94.

viii. Lumen maintenance of 70% for at least 50000 hours for Interior and exterior applications. Data sheet showing lumen maintenance in the LED shall also be submitted.

ix. The LED should have following qualities:

a. A unique, patented system of positioning LEDs in order to achieve optimal even light distribution.
b. An embedded intelligent driver allowing a configuration which serves the needs of a particular installation.
c. A built in thermal protection from overheating of the source of light.
d. The source of light safe for the environment.
e. High mechanical and vibration resistance of LEDs

Qualities of the intelligent controller:

a. Soft Start function – smooth increase of brightness from 0% to 100% in programmable time.
b. Electronic compensation of environment temperature influence on luminous flux value.
c. Optional Aging Compensation function – Electronic compensation of source of light ageing influence on luminous flux value through increasing the luminaire power in time.

LED should have the following Features:

1. Intelligent Current control
2. No Adverse Glare
3. No light pollution
4. No High-voltage non dust absorption
5. No High-temperature, No aging Yellow lampshade
6. Start without delay
7. No strobe
8. Impact Resistance, Shock-proof, without ultraviolet (UV) and infrared (IR) Radiation
9. High color index,
10. High luminous efficiency

POWER ELECTRONICS:

1. The components used in the electronics circuit should be RoHS compliant and the certificates for the same should be made available.

2. The connecting wires used in the circuit should be of UL grade and the certificate of the same shall be made available on request.

3. constant current out put for wide input voltage range.
4. MCPCB should possess thermal conductivity of up to 3W/m-^{0} K
5. MCPCB thermal resistance should be less than 0.35^0 C-cm2/watt.

Protection

1. over current
2. over voltage
3. over temperature
4. under voltage protection
5. out put short circuit protection

Testing procedures and certificates

1. Luminaire Photometry reports by NABL/DOE certified laboratories.
2. RoHS compliance certificate for all electronics components used.
3. The IES files of the Luminaires should be made available on request.
4. Life expectancy test of LEDs.

In- House Manufacturing, Assembly, Testing and Demonstration of product.

1. Surface Temperatures Measurements
2. Over voltage protection
3. over current protection
4. over temperature protection
5. Humidity test
6. Insulation resistance, electrical strength and leakage current
7. Creepage distance and clearance
8. LED Mounting on PCBs
9. Electrical circuit assembly
10. Pre dispatch tests conducted as per IEC/IS

3.19 **DRAWINGS:**

3.19.1 The manufacturer shall supply four copies of the drawings which will describe the equipment in detail for approval. After approval six complete sets of final drawings for each substation shall be sent. The reproducible drawings may also be supplied.

3.19.2 The following drawings for each item are to be supplied as a part of the contract.

a) Outline dimensional drawings of LED Luminaries / Fittings and accessories.
a) Assembly drawings and weight of main components.
c) Test reports (for both type and routine tests).

3.19.3 Six copies of instruction booklets/operation and maintenance manuals shall be supplied for each substation. Descriptive literature and data on LED Luminaries / Fittings, construction, shall be supplied for each LED Luminaries / Fittings.

3.20 GUARANTEED TECHNICAL AND OTHER PARTICULARS:

The Guaranteed technical and other particulars of the LED Luminaries/Fittings shall be furnished in the proforma given in Annexure-I.

4.0 TESTS

a) All the offered equipment shall be fully type tested by the bidder as per the relevant standard. The type tests must have been conducted on the same rating of LED Luminaries/Fittings not earlier than ten years as on the date of bid opening. The bidder shall furnish two sets of the following type test reports as per relevant standards for each type of equipment offered along with the bid. The offers received without these test reports shall be treated as NON-RESPONSIVE .

b) Special tests other than type and routine tests, Bidder shall also be carry out any tests that are required to establish the performance of the Light fittings offered as per the technical specification at any stage at the bidder's cost.

c) The pre-shipment checks to be carried out by the Bidder are given under clause 6.0, "Inspection".

d) The bidder shall offer the LED Luminaries / Fittings type tested. The type tests should have been conducted not earlier than 10 years as on date of bid opening. In the absence of not having valid type test reports as above, the successful bidder shall conduct the type tests on the offered equipment before supply.

5.1 Routine tests :

All standard routine tests in accordance with applicable International /Indian Standards and latest amendments shall be carried out on the LED Luminaries / Fittings.

Terminals connectors shall be tested as per IS: 5561. In addition, the Bidder shall carry out an additional test to demonstrate withstand capacity of wind load. The test procedure and acceptance norms shall be furnished for Purchaser's approval.

2.1 Test reports

a) Four (4) sets of certified test reports and oscillograms shall be submitted for approval prior to despatch of the equipment. The equipment shall be dispatched only when all the required type and routine tests have been carried out and test reports have been approved by the Purchaser.

b) Four (4) copies of the test reports for the tests carried out on the ancillary apparatus shall be furnished to the Purchaser for approval prior to despatch.

c) All auxiliary equipment shall be tested as per the relevant standard. Test certificate shall be submitted for bought out items.

6.0 INSPECTION

6.1 General

i) The owner shall have access at all times to the works and all other places of manufacture where the LED Luminaries/Fittings are being manufactured and the Bidder shall provide all facilities for unrestricted inspection of the Bidder's works, raw materials, manufacture of all the accessories and for conducting necessary tests as detailed herein.

ii) The successful Bidder shall keep the Purchaser informed in advance of the time of starting and of the progress of manufacture of equipment in its various stages so that arrangements could be made for inspection.

iii) No material shall be dispatched from its point of manufacture unless the material has been satisfactorily inspected and tested.

iv) The acceptance of any quantity of equipment shall in no way relieve the successful Bidder of his responsibility for meeting all the requirement of this specification and shall not prevent subsequent rejection if such equipment are later found to be defective.

6.2 Inspection programme

a) The Bidder shall chalk out a detailed inspection and testing programme for manufacturing activities for the various components. An indicative programme of inspection as envisaged by the Purchaser. This is not however intended to form a comprehensive programme as it is Bidder's responsibility to draw up and carry out such a programme duly approved by the Purchaser. The owner reserves the right to carryout any tests by a third party.

b) Cost of inspection/tests and third party inspection if any shall be borne by the Bidder.

c) Additional tests, if required, are deemed to be included in the scope of work.

d) Stages of inspection and Purchaser's participation would be defined and tied up at the time of award of contract.

7.0 QUALITY ASSURANCE PLAN :

7.1 The bidder shall invariably furnish the following information along with his offer failing which the offer shall be liable for rejection. Information shall be separately given for individual type of equipment offered.

a) The Structure of organization

b) The duties and responsibilities assigned to staff ensuring quality of work

c) The system of purchasing, taking delivery and verification of materials

d) The system for ensuring quality of workmanship

e) The quality assurance arrangements shall confirm to the relevant requirement of ISO-9001 on ISO 9002 as applicable.

f) Statement giving list of important raw materials, names of sub-supplies for the raw materials, list of standards according to which the raw material are tested, list of tests normally carried out on raw material in the presence of suppliers representative, copies of test certificates.

g) Information and copies of test certificates as on (i) above in respect of bought out items.

h) List of manufacturing facilities available

i) Level of automation achieved and list of areas where manual processing exists.

j) List of areas in manufacturing process, where stage inspections are normally carried out for quality control and details of such tests and inspections.

k) List of testing equipment available with the bidder for final testing of equipment specified and test plant limitation, if any vis-a-vis the type. Special acceptance and routine tests specified in the relevant standards. These limitations shall be very clearly brought out in "Schedule of Deviations" from the specified test requirements.

7.2 The Contractor shall within 30 days of placement of order, submit the following information to the purchaser.

i) List of the raw material as well as bought out accessories and the names of sub-suppliers selected from those furnished along with the offer.

ii) Type test certificates of the raw material and bought out accessories if required by the purchaser.

iii) Quality Assurance Plan (QAP) with hold points for purchasers inspection. QAP and purchasers hold points shall be discussed between the purchaser and contractor before the QAP is finalised.

The contractor shall submit the routine test certificates of bought out accessories and central excise invoice for raw material at the time of routine testing if required by the purchaser and ensure that the quality assurance requirements of specification are followed by the sub-contractor.

7.3.1 The Quality Assurance Programme shall give a description of the Quality System and Quality Plans with the following details.

a) Quality System:

* The Structure of the Organisation
* The duties and responsibilities assigned to staff ensuring quality of work.
* The system of purchasing, taking delivery of verification of materials.
* The system of ensuring of quality workmanship.
* The system of control of documentation.
* The system of retention of records.
* The arrangement of contractor internal auditing.
* A list of Administrator and work procedures required to achieve contractors quality requirements. These procedures shall be made readily available to the purchaser for inspection on request.

b) Quality Plans:

a. An outline of the proposed work and program sequence.
b. The structure of contractors organizations for the contract.
c. The duties and responsibilities ensuring quality of work
d. Hold and notification points.
e. Submission of engineering documents required by this specification.
f. The Inspection of the materials and components on request.
g. Reference to contractors work procedures appropriate to each activity.
h. Inspection during fabrication/construction.
i. Final inspection and test.

8.0 DOCUMENTATION:

8.1 i) The Bidder shall furnish along with the bid the dimensional drawings of LED Luminaries / Fittings, GA and all other accessories.

ii) These drawings shall include the following information.

a) Detailed internal drawings of LED Luminaries / Fittings clearly indicating all the details of assembly incorporating all important dimensions, ratings etc.

b) Tolerances on dimensions.

c) Material designation used for different components with reference to standards.

d) Fabrication details such as welds, finishes and coatings.

e) Catalogue or part numbers for each component and the total assembly with bill of materials.

f) Identification marking.

g) Weight of individual components and total assembled weight.

8.2 **Test Reports**

i) Four copies of type test reports shall be furnished to the Purchaser within one month of conducting the tests. one copy will be returned duly certified by Purchaser to the Bidder within three weeks thereafter and on receipt of the same Bidder shall commence with the commercial production of the concerned material.

ii) Four copies of acceptance test reports shall be furnished to the Purchaser. One copy will be returned duly certified by the Purchaser and only thereafter shall the materials be dispatched.

iii) All records of routine test reports shall be maintained by the Bidder at his works for periodic inspection by the Purchaser.

iv) All test reports for tests conducted during manufacture shall be maintained by the Bidder. These shall be produced for verification as and when requested for by the Purchaser.

8.3 All drawings shall conform to International Standards Organization (ISO) 'A' series of drawing sheet/Indian Standards Specification. All drawings shall be in ink and suitable for micro filming. All dimensions and data shall be in S.I. Units only.

The Bidder shall within 2 weeks of placement of order, submit four sets of final versions of all the above drawings for Purchaser's approval. The Purchaser shall communicate his comments/approval on the drawings to the Bidder within reasonable time. The Bidder shall, if necessary, modify the drawings and resubmit four copies of the modified drawings for Purchaser's approval within two weeks from the date of Purchaser's comments.

After receipt of Purchaser's approval, the Bidder shall within three weeks, submit 6 prints and two good quality reproducible of the approved drawings for Purchaser's use.

8.4 Adequate copies of contract drawings, acceptance and routine test certificates, duly approved by the Purchaser shall accompany each of the dispatched consignment.

8.5 The manufacturing of the equipment shall be strictly in accordance with the approved drawings and no deviation shall be permitted without the written approval of the Purchaser. All manufacturing and fabrication work in connection with the equipment prior to the approval of the drawing shall be at the Bidder's risk only.

8.6 Six copies of nicely printed and bound volumes of operation, maintenance and erection manuals in English language shall be supplied by the Bidder along with each LED Luminaries / Fittings to destination. The manual shall contain all the drawings and information required for erection, operation and maintenance of

theLED Luminaries/Fittings. The manual shall also contain a set of all the approved drawings, type, routine & acceptance Test reports etc.

8.6 Approval of drawings by the Purchaser shall not relieve the Bidder of his responsibility and liability for ensuring correctness and correct interpretation of the latest revision of applicable standards, rules and codes of practices. The equipment shall conform in all respects to high standards of engineering, design, workmanship and latest revisions of relevant standards at the time of ordering and the Purchaser shall have the power to reject any material which in his judgment is not in full accordance therewith.

9 PACKING AND FORWARDING:

9.1 The equipment shall be packed in crates suitable for vertical/horizontal transport as the case may be and suitable to withstand handling during transport and outdoor storage during transit. The Bidder shall be responsible for any damage to the equipment during transit, due to improper and inadequate packing. The easily damageable material shall be carefully packed and marked with the appropriate caution symbol. Wherever necessary, proper arrangement for lifting, such as lifting hooks etc. shall be provided. Any material found short inside the packing cases shall be supplied by Bidder without any extra cost.

9.2 Each consignment shall be accompanied by a detailed packing list containing the following information:

a) Name of the consignee
b) Details of consignment
c) Destination
d) Total weight of consignment
e) Handling and unpacking instructions.
f) Bill of material indicating contents of each package.

The Bidder shall ensure that the packing and bill of material are approved by the Purchaser before despatch.

9.3 Weight of the LED Luminaries / Fittings shall be designed, so that the LED Luminaries / Fittings, can be transported by road. Heaviest single packet that can be transported by owner's trailor is about 60 MTs.

10.0 QUANTITY AND DELIVERY REQUIREMENTS :

i) This is set out in Schedule.
ii) The scope of supply shall include a supply of 2.5% extra quantity of bolts, nuts, washers, split pins, cotter pins and such other small loose items free of cost.

ANNEXURE-1
GUARANTEED TECHNICAL PARTICULARS
(TO BE SUBMITTED BY THE BIDDER)

Sl. No.	Parameters	Details/Guaranteed value
1.	Name of the manufacturer of the Light fitting offered Complete Postal address with Fax And Phone numbers.	
2.	Name of the manufacturer of the LEDs that go in to the light fitting offered with Complete Postal address with Fax And Phone numbers	
3.	Names of the manufacturers of all the other components that go in to the light fitting offered with Complete Postal addresses with Fax And Phone numbers of all these manufacturers.	
	LUMINAIRE	
4.	Output Luminous Flux of luminaire	
5.	Average Lux Level of luminaire	
6.	Beam Angle of luminaire	
7.	Uniformity	
8.	Photometric Curve	
9.	Material of Luminaire & Diffuser	
10.	Dimensions	
11.	Weight of the luminaire	
12.	Operating temperature	
13.	Adjusting of the luminaire inclination	
14.	Luminaire finishing colours	
15.	Body & lamp shade material	
	Working temperature.	
17.	Working humidity	
18.	Working life	
19.	Lamp's efficiency	
20.	Fixture efficiency	
21.	Light Distribution curve /beam pattern	
22.	Lamp base	
23.	The highest light intensity angle	
24.	Electrical insulation class	
25.	Conformity with IP-65 fixtures	

Sl. No.	Parameters	Details/Guaranteed value
26.	Conformity with UL94	
27.	Surface temperature of Luminaire	
28.	Power Consumption of Luminaire	
29.	IEC/IS applicable	
30.	Illumination (E) of LED luminaire equal to HPSV lamps for the heights 6m, 8m, 10m & 12m.	
31.	Effective Illuminated area from the height of 6m, 8m, 10m & 12m.	
	LED	
32.	LED Operating Current	
33.	LED Life	
34.	LED power consumption	
35.	CCT (Correlated Colour Temperature)	
36.	Color Rendering Index (CRI)	
37.	LED Junction temperature.	
38.	LED Beam angle	
	MCPCB	
39.	Operating temperature	
40.	Junction temperature of Electronic components	
41.	System resistance	
42.	Heat dissipation & cooling process	
43.	Thermal conductivity	
44.	Thermal resistance	
45.	IEC applicable	
	Electrical	
46.	Input voltage	
47.	Power Factor	
48.	Input frequency	
49.	Driver efficiency	
50.	THD (Total Harmonic Distortion)	